

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
QUADRATIC EQUATIONS | LEVEL 4*Previous-year questions – AIEEE and JEE (Main), year tagged.*

1. If the roots of the equation $bx^2 + cx + a = 0$ are imaginary, then for all real values of x the expression $3b^2x^2 + 6bcx + 2c^2$ is [AIEEE 2009]
(A) greater than $-4ab$ (B) greater than $4ab$
(C) less than $4ab$ (D) less than $-4ab$
2. Let $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then $p(2)$ is [AIEEE 2011]
(A) 9 (B) 3
(C) 18 (D) 6
3. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up with the roots $(4, 3)$. Rahul made a mistake in writing down the coefficient of x and got the roots $(3, 2)$. The correct roots of the equation are [AIEEE 2011]
(A) $-4, -3$ (B) $6, 1$
(C) $4, 3$ (D) $-6, -1$
4. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is [JEE (Main) 2013]
(A) $3 : 2 : 1$ (B) $3 : 1 : 2$
(C) $1 : 3 : 2$ (D) $1 : 2 : 3$
5. The sum of all real values of x satisfying $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is [JEE (Main) 2016]
(A) 6 (B) -4
(C) 3 (D) 5
6. Let $S = \{x \in \mathbb{R} : x \geq 0 \text{ and } 2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$. Then S [JEE (Main) 2018]
(A) contains exactly one element
(B) contains exactly two elements
(C) is an empty set
(D) contains exactly four elements
7. The number of positive integral values of α for which the roots of $6x^2 - 11x + \alpha = 0$ are rational numbers is [JEE (Main) 2019]

- (A) 5 (B) 3
(C) 4 (D) 2

8. The value of λ for which the sum of the squares of the roots of $x^2 + (3 - \lambda)x + 2 = \lambda$ is least is [JEE (Main) 2019]

- (A) 1 (B) $\frac{4}{9}$
(C) 2 (D) $\frac{15}{8}$

9. If one real root of $81x^2 + kx + 256 = 0$ is the cube of the other root, then a value of k is [JEE (Main) 2019]

- (A) -81 (B) -300
(C) 144 (D) 100

10. If λ is the ratio of the roots of $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$ is [JEE (Main) 2019]

- (A) $4 - 3\sqrt{2}$ (B) $4 - 2\sqrt{3}$
(C) $-2 + \sqrt{2}$ (D) $2 - \sqrt{3}$

11. The number of integral values of m for which the quadratic expression $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, $x \in \mathbb{R}$, is always positive, is [JEE (Main) 2019]

- (A) 3 (B) 7
(C) 8 (D) 6

12. The sum of the solutions of $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$, $x > 0$, is [JEE (Main) 2019]

- (A) 10 (B) 4
(C) 9 (D) 12

13. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following is correct? [JEE (Main) 2019]

- (A) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
(B) d, e, f are in G.P.
(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.
(D) d, e, f are in A.P.

14. If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ so that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is [JEE (Main) 2019]

- (A) $10\sqrt{5}$ (B) $8\sqrt{3}$
(C) $8\sqrt{5}$ (D) $4\sqrt{3}$

15. Let S be the set of all real roots of $3^x (3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S [JEE (Main) 2020]
(A) contains at least four elements
(B) contains exactly two elements
(C) is an empty set
(D) is a singleton
16. Let $a, b \in \mathbb{R}$, $a \neq 0$, be such that $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is [JEE (Main) 2020]
(A) 26 (B) 24
(C) 25 (D) 28
17. Let $f(x)$ be a quadratic polynomial with $f(-1) + f(2) = 0$. If one root of $f(x) = 0$ is 3, then its other root lies in [JEE (Main) 2020]
(A) $(-1, 0)$ (B) $(0, 1)$
(C) $(1, 3)$ (D) $(-3, -1)$
18. If α, β are the roots of $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right) \left(\beta - \frac{1}{\beta}\right) \left(\alpha + \frac{1}{\beta}\right) \left(\beta + \frac{1}{\alpha}\right)$ equals [JEE (Main) 2020]
(A) $\frac{9}{4} (9 + q^2)$ (B) $\frac{9}{4} (9 + p^2)$
(C) $\frac{9}{4} (9 - q^2)$ (D) $\frac{9}{4} (9 - p^2)$
19. Let $\lambda \neq 0$ be real. If α, β are the roots of $x^2 - x + 2\lambda = 0$ and α, γ are the roots of $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is [JEE (Main) 2020]
(A) 27 (B) 9
(C) 36 (D) 18
20. The product of the roots of $9x^2 - 18|x| + 5 = 0$ is [JEE (Main) 2020]
(A) $\frac{25}{9}$ (B) $\frac{5}{9}$
(C) $\frac{5}{27}$ (D) $\frac{25}{81}$
21. If α, β are the roots of $7x^2 - 3x - 2 = 0$, then $\frac{\alpha}{1 - \alpha^2} + \frac{\beta}{1 - \beta^2}$ equals [JEE (Main) 2020]
(A) $\frac{27}{16}$ (B) $\frac{27}{32}$
(C) $\frac{1}{24}$ (D) $\frac{3}{8}$

22. If α, β are the roots of $x^2 - 64x + 256 = 0$, then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$ is **[JEE (Main) 2020]**
- (A) 1 (B) 4
(C) 3 (D) 2
23. If α and β are the roots of $2x(2x + 1) = 1$, then β is equal to **[JEE (Main) 2020]**
- (A) $2\alpha^2$ (B) $-2\alpha(\alpha + 1)$
(C) $2\alpha(\alpha + 1)$ (D) $2\alpha(\alpha - 1)$
24. The least positive value of a for which $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is **[JEE (Main) 2020]**
- (A) 8 (B) 12
(C) 6 (D) 10
25. The integer k for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every $x \in \mathbb{R}$ is **[JEE (Main) 2021]**
- (A) 3 (B) 0
(C) 2 (D) 4